A Generic Simulation Framework for Intra-Warehouse Operations with Automatic Model Generation

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Abstract
Inside a warehouse, managers decide where to place storage and pickup areas, how to route material flow, how many forklifts and workers to run and which picking and loading strategies to follow, usually by rules of thumb. Each new layout normally needs a new simulation model, which is slow and requires an expert. We want to build one generic, parameter driven discrete event model of a warehouse, covering receiving, putaway, storage, order picking, consolidation, staging and loading, that a non-expert can configure and run. On top of this framework we plan to develop an automatic model generation framework: the user draws a facility on a grid of standard building blocks and the complete simulation model is generated from it. The framework is used to size forklifts and workers, compare storage and pickup allotments and test operating strategies under synthetic stochastic demand.

1. System Selection & Problem Statement

The System

The internal operations of a warehouse. Inbound pallets arrive at the docks, are checked in a receiving buffer and put away to storage by forklifts. Customer orders release picking tasks; picked items are consolidated, held in pickup areas and loaded at outbound docks. Trucks appear only as the arrival and departure points at the dock. The model is built in three components:

- **User Interface:** a non-expert sets layout, docks, forklifts, volume and policies, launches experiments and reads a KPI dashboard.
- **Generic Modelling Framework:** reusable building blocks (Dock, Forklift/charging, Passage), generic process logic and generic agents for pallets, orders, forklifts and workers.
- **Data & Model Assembly:** stores configurations and input distributions, assembles the runnable model and exports results.

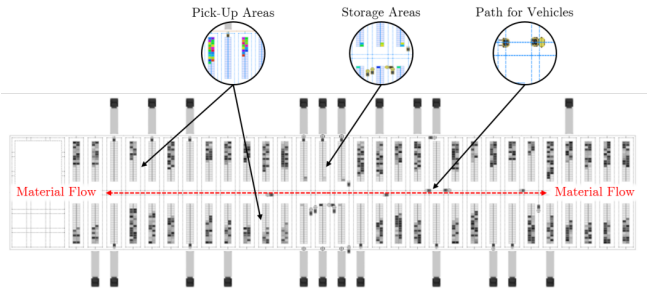


Figure 1. Simulation Model

Highlight: Automatic Model Generation

Building a new AnyLogic model for every layout is the main cost of simulation-based layout planning. We want to generate it automatically. The user fills a grid matrix where each cell is a building block (D = dock block, F = forklift block, P = passage, 0 = empty). A generator reads the matrix and creates the space markup, walls, docks, storage and pickup areas and the forklift path network at runtime. Any facility shape (I, L, T, U, H) then becomes a one-click experiment instead of a modelling task. This is built as an extension on top of the working framework.

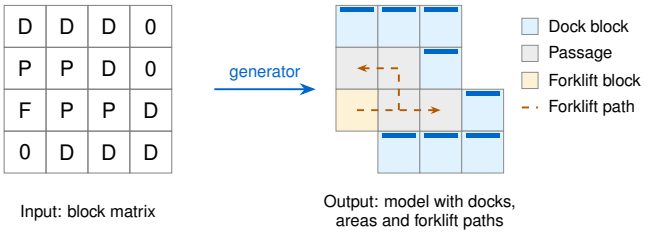


Figure 2. Automatic model generation: a block matrix is converted into a complete, runnable facility model.

The Problem

Inside the warehouse, poorly allotted storage and pickup areas, long or conflicting forklift paths and unbalanced forklift and worker capacity cause excess travel, congestion in aisles and buffers, and orders that miss their loading slot. We study:

1. **Allotment & material flow:** where should storage and pickup areas sit relative to the docks, and how should forklift paths be routed, to minimise travel distance and shipment cycle time?
2. **Resources:** what is the minimum number of forklifts and workers that meets the service level at peak volume, and which resource is the bottleneck?
3. **Strategies:** how much do storage assignment, picking and loading strategies improve performance at zero capital

cost?

4. **Usability:** can one generic model, and ultimately one generated from a drawing, answer these without manual rebuilding?

System Boundaries

Included	Excluded
Dock unloading and loading	Yard, gate and truck scheduling
Receiving check and buffer	Road transport outside the facility
Putaway and storage	Procurement and inventory policy
Picking, consolidation, staging	Value-added services
Forklift travel, congestion, charging	Shifts, breaks, labour rules
Worker tasks and utilisation	Returns, damages, IT systems

2. System Configuration

Entities & Resources

Entity	Attributes
Inbound pallet	Arrival time, SKU class, storage location
Customer order	Lines per order, due loading slot
Shipment	Destination, pallets, loading dock
Forklift task	Type (putaway, pick, load), origin, destination
Resource	Capacity / decision
Forklifts	Fleet size; performance
Workers	Receivers, pickers, loaders per shift
Docks	Count; inbound / outbound split
Receiving buffer	Pallet capacity
Storage area	Slots; placement; SKU zoning
Pickup area	Pallet capacity; placement
Aisles / paths	One-way or two-way; congestion
Chargers	Count

Process Flow Sketch

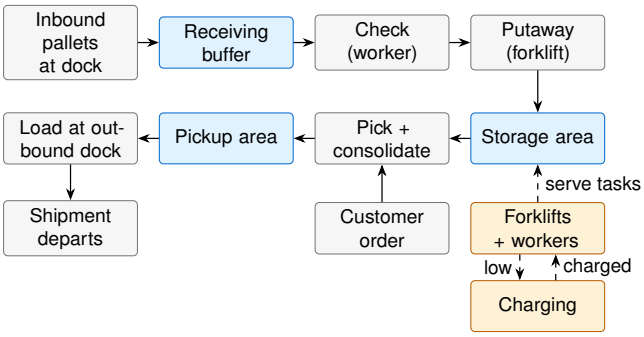


Figure 3. Intra-warehouse routing logic. Blue: buffers and areas with finite capacity; orange: shared forklift and worker resources.

3. Input Data Collection Plan

All inputs use synthetic data. Distributions are fitted in Python, checked with KS / Chi-square tests, and the model is run on both sampled data and fitted distributions.

Stochastic variable	Sourcing	Sample / Dist.
Inbound pallet arrivals at dock	Synthetic	500; Poisson
Order inter-arrival	Synthetic	1000; Exponential
Lines per order	Synthetic	1000; Geometric
SKU demand mix	Synthetic	ABC (skewed)
Receiving check time	Synthetic	500; Gamma
Pick / consolidation time	Synthetic	500; Lognormal
Unload / load time per pallet	Synthetic	500; Lognormal
Forklift speed, charge time	Synthetic	Normal / Uniform
Forklift MTBF / MTTR	Synthetic	200; Exp. / Lognormal

4. Output Measures & Optimization Goal

Key Performance Indicators

Category	KPI
Time	Shipment cycle time (mean, p95), forklift cycle time
Volume	Handling volume, system load
Utilisation	Forklifts, workers, docks, storage and pickup areas (%)
Distance	Distance travelled by forklifts
Environment	Carbon footprint (forklift energy use)
Congestion	Buffer overflow, aisle blocking time

The "What If" / Optimization Goal

Input parameter	Levels
Number of forklifts	Minimum feasible to saturation
Number of workers	Per role: receiving, picking, loading
Forklift performance	Standard vs. high-speed
Number of docks	12, 18, 24 (inbound / outbound split)
Picking strategy	Single order, batch, zone
Loading strategy	FIFO, by destination, by due slot
Shipment volume	Baseline, +20%, +40%
Design criterion	Levels
Layout form	I, L, T; any shape via generation
Storage allotment	Random, ABC near docks, SKU zones
Pickup allotment	Near outbound docks, centralised
Material flow, paths	One-way, two-way aisles
Buffer capacity	Low, medium, high

Optimization goal (OptQuest): minimise the number of forklifts and workers subject to p95 shipment cycle time $\leq T^*$ and zero buffer overflow; then minimise forklift travel distance and carbon footprint through area allotment, path design and strategy choice.

References

[1] L. Jurgeleit, M. Mowe, M. Kiefer, C. Schumacher, and U. Clausen. Automatic model generation for discrete event simulation of less-than-truckload terminals. *Proc. Winter Simulation Conference*, 1493–1504, 2024.